

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

S2 B.Tech CSEAI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – 2 (Design)

Architectural Modelling (CO2)

Group No : 23

Team Members :

Roll No	Student Name
AM.SC.UAIE25107	C V Rohith Reddy
AM.SC.UAIE25130	M Pramod
AM.SC.UAIE25132	M Srirag Sarma
AM.SC.UAIE25135	N Charan

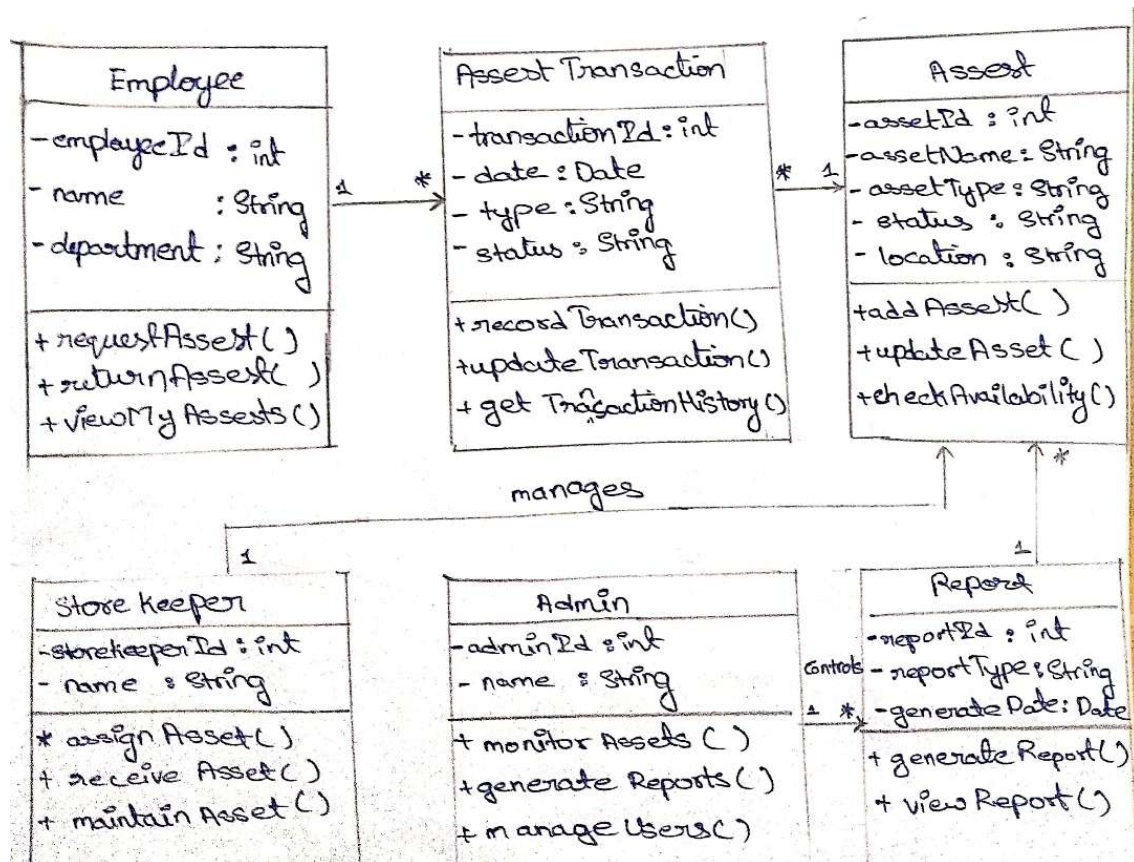
Problem Title : **ASSET TRACKING SYSTEM**

STRUCTURAL DESIGN

1. Class diagram .

The class diagram represents the structure of the Asset Tracking System by showing different classes, their attributes, methods, and relationships between them. The main classes include Asset, Employee, Storekeeper, Admin, AssetTransaction, and Report.

The Employee interacts with Asset through AssetTransaction, while the Storekeeper manages asset allocation. The Admin oversees the system and generates reports. Each class is connected logically to reflect real-world interactions.



BEHAVIUIORAL DESIGN

2. Use-case diagram

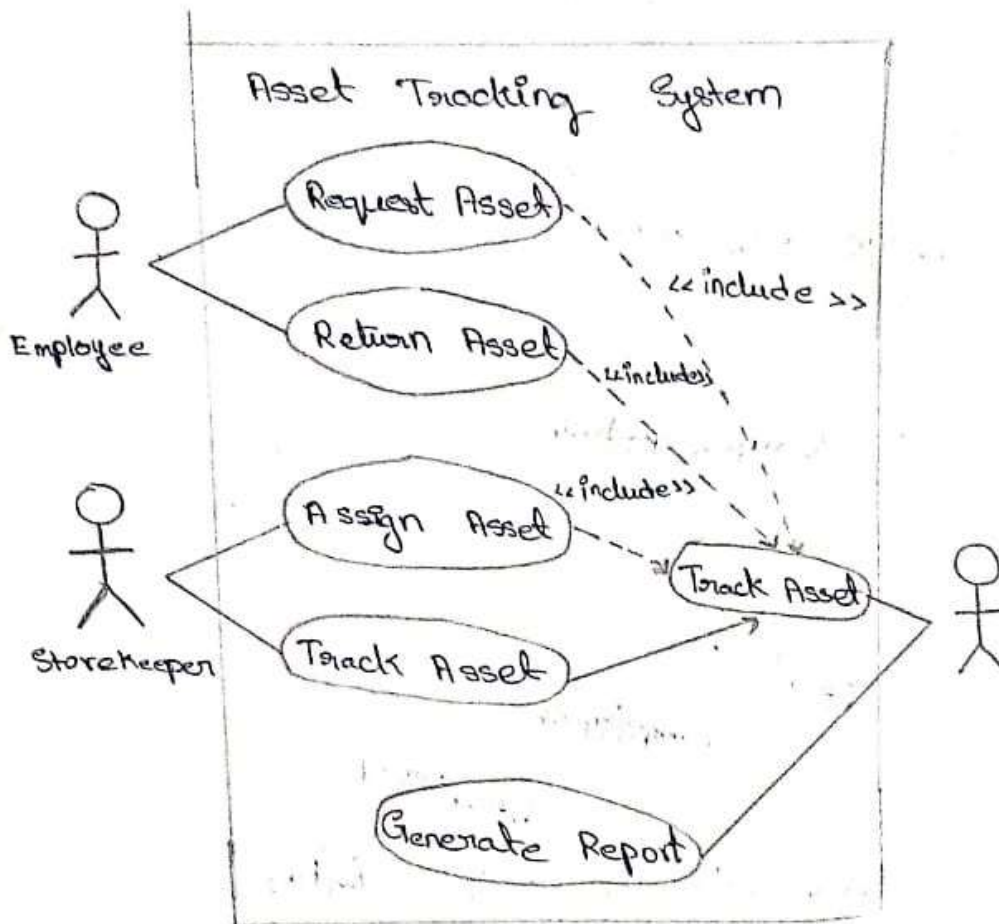
a. List of actors

1. Employee
2. Storekeeper
3. Admin

b. List of use-cases

1. *Request Asset – Employee requests an asset from the system.*
2. *Assign Asset – Storekeeper assigns assets to employees.*
3. *Return Asset – Employee returns the asset after use.*
4. *Track Asset – System tracks asset status and location.*
5. *Generate Report – Admin generates reports for monitoring.*

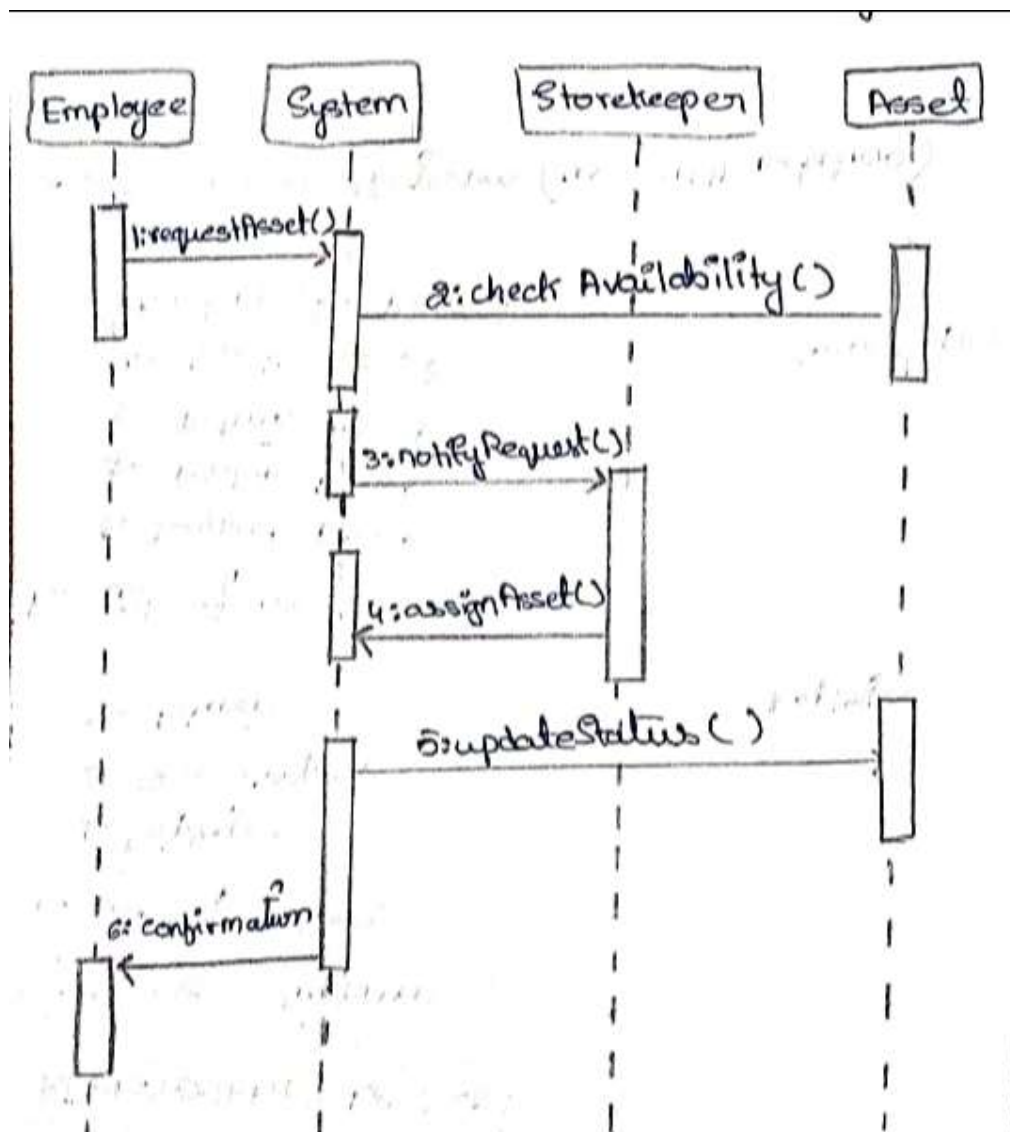
c. Use case diagram - with UML notations



3. **Sequence diagram** - with UML notations

- a. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.

An employee requests an asset through the system. The system checks availability and forwards the request to the storekeeper. The storekeeper assigns the asset, and the system updates the transaction details.



- b. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.

An employee returns an asset to the system. The storekeeper receives the asset, and the system updates its availability and records the transaction.

